

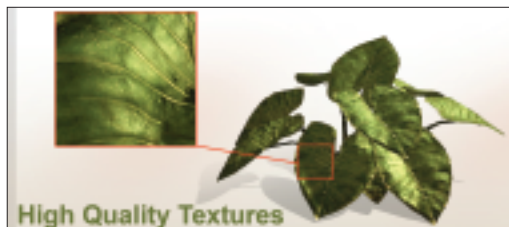
236 Watts! However, NVIDIA has taken a holistic approach to power management and the actual thermal profile of the chip will vary greatly depending what it will be used for.

Broadly, the chip operates under four different thermal envelopes:

- Idle/2D power mode (approx 25 W)
- Blu-ray DVD playback mode (approx 35 W)
- Full 3D performance mode (up to 236 W)
- HybridPower mode (effectively 0 W)

The HybridPower mode is enabled when a GeForce card is used in conjunction with special nForce motherboards: in this case, the onboard graphics chipset is used for 2D tasks and the dedicated graphics card is effectively switched off.

While HybridPower might be considered a feature outside of the GTX200—the chip itself has several tricks up it sleeve to reduce power draw. For one, when running 3D applications, the driver can seamlessly switch between power modes based on GPU utilisation. These new



Features of future games such as Far Cry 2 will demand an increase in the number and size of textures. The GTX 200 chip can support up to 1 GB video card memory

chips come with activity monitors which act as watchdogs over the traffic inside the GPU. Based on the activity reported, the GPU driver can dynamically throttle the performance mode by making changes to the clock speed and the voltage level—all transparent to the user. This behind-the-scenes speed throttling goes a long way to supplying just the needed juice to the GPU and keeping the temperature and the power draw down. Apart from this throttling, the GPU also has clock-gating circuitry, which effectively shuts down blocks of the GPU which are not being used at a particular time, further reducing power during periods of non-peak GPU utilisation.

Compute This

NVIDIA wants to take its GPU outside the graphics processing threshold. It has already taken steps to that end—the largest being the intro-

duction of its C-like CUDA interface for programmers. Where NVIDIA would like to see its GPU (and ATI too, as a matter of fact) is more general computing. To this end some programs have already been showcased—traditional tasks which are either accelerated greatly or completely executed on the GPU. Adobe, for example, showcased a version of its Photoshop application which was GPU-accelerated and was capable of quickly loading and editing a 442MP file, at 2GB size. The edits done were rudimentary—zooming in and out, rotation and so on. But as a taste of things to come it was an impressive display, given how slow Photoshop can get while manipulating large images. Another piece of software used the GPU to encode video files faster than real-time playback, yet another was a ray-tracing application which made use of the GPU. Other tasks the GPU is being called for include protein folding via a Folding@Home client and physics simulations for liquids, soft bodies, and fabrics.

Compete This

There are to be two variants based on the GTX 200 chip—the GeForce GTX 280 will be the higher-end part (priced at around \$650), and the GeForce GTX 260 will lose some of its components and be priced slightly lower (at about \$399). Early benchmarks show the GTX 280 to be on par with the 9800 X2 cards on some tests, and ahead of the X2 in others. Going on the data that's out there, NVIDIA might just lose its price to performance crown to its rival's graphic chipset. ATI/AMD has taken a completely different approach to designing its graphics part. No technical information is available to make an objective analysis between the two approaches, but from a holistic point-of-view: while NVIDIA has taken the bigger, more powerful approach; ATI has gone down the smaller, scalable route. ATI's new chip—the RV770 seems designed to run in pairs and efficiently. Early tests put the RV770-based, single chip solution the Radeon HD 4850 within range of the 9800GTX, and at \$200 or less. The 4850 in crossfire configuration gets within touching distance of the GeForce GTX 280, and at \$100 less. AMD/ATI will also release a \$300 card the Radeon HD 4870 which will run two RV770 chips on the same graphics card and will come equipped with GDDR5 memory—about 80 per cent faster than GDDR3 used in the GeForce cards.

One thing's for sure—things are certainly heating up in the graphics market. At one end you have the ATI and NVIDIA rivalry, which has already lead to price drops from the NVIDIA camp; on the other hand you have the upcoming challenger to the market—Intel. Who knows what that will bring to the scene? Well, cheaper prices for sure. This is exciting times for us consumers indeed! ☒

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